

WANTED

ILL-POSED AUTOREGRESSIVE PROBLEMS

AUTOREGRESSIVE MODELS

AR(P) with a Gaussian prior on the coefficients is conjugate: the posterior stays closed form.

$$x_t = a_1 x_{t-1} + \dots + a_P x_{t-P} + \varepsilon_t$$

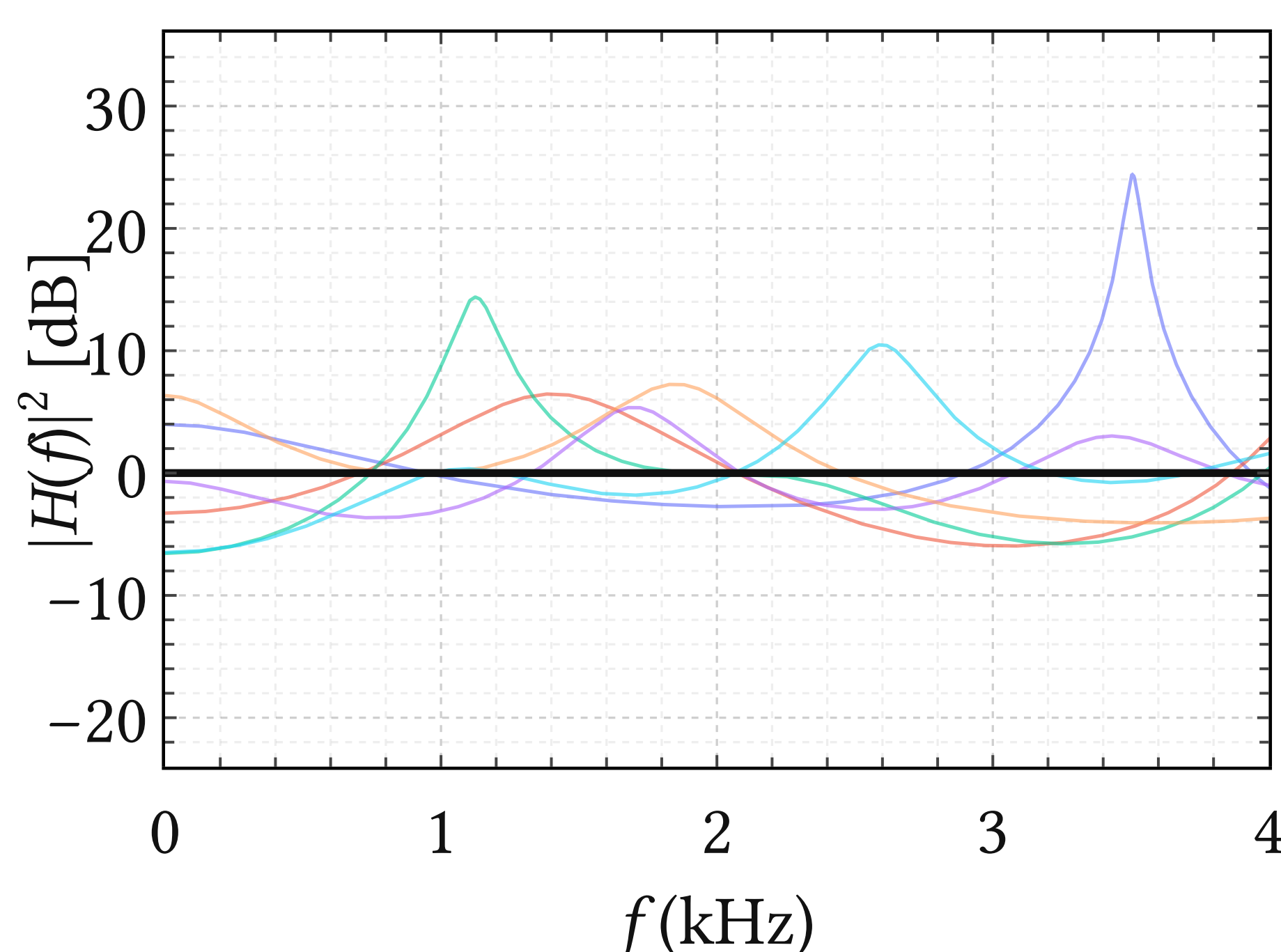
$$\mathbf{a} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{S})$$

The filter $H(z) = 1/A(z)$ is set by the polynomial $A(z) = 1 - a_1 z^{-1} - \dots - a_P z^{-P}$, and polynomials factor:

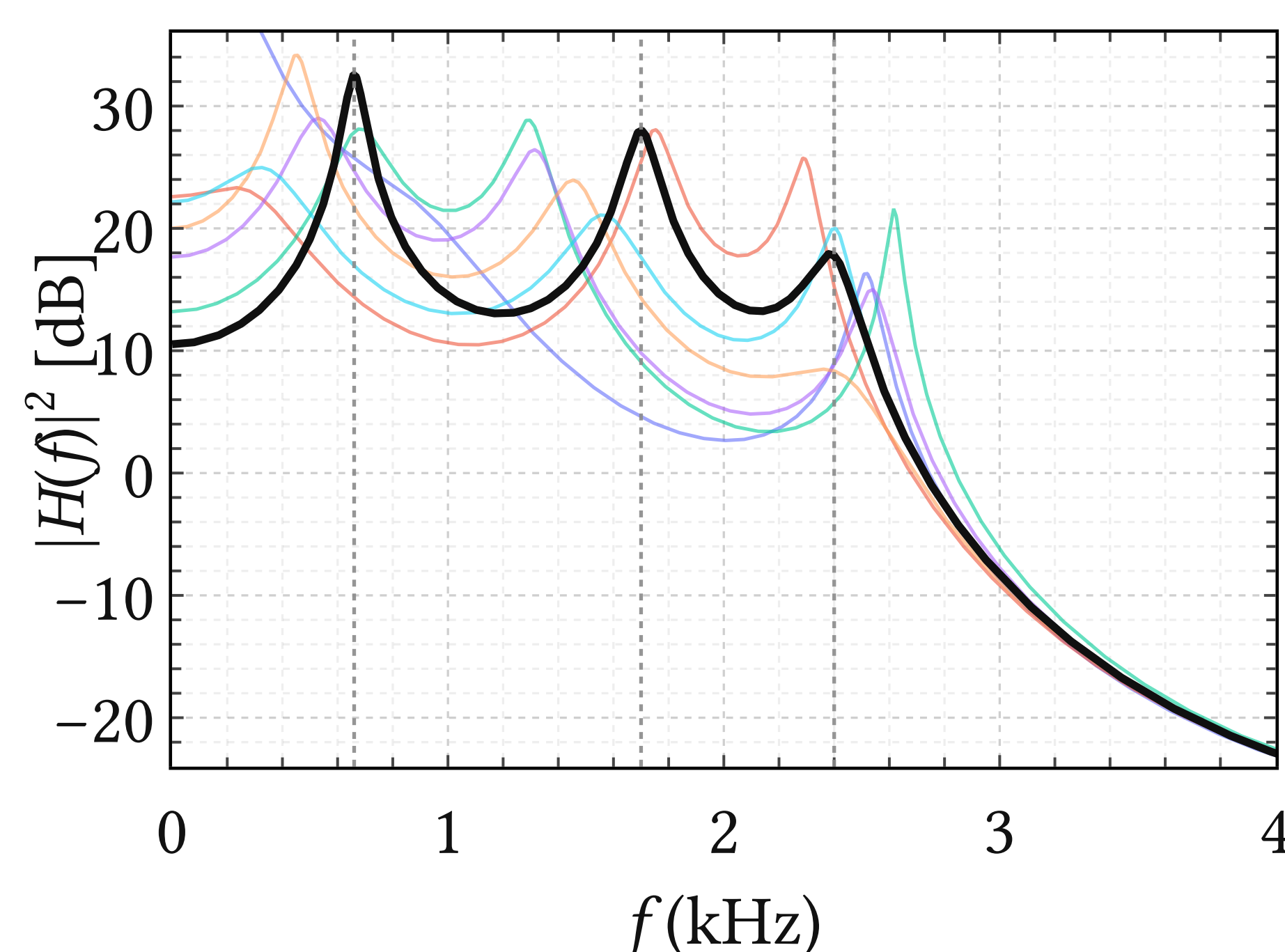
$$A = \underbrace{Q_1 \dots Q_K}_{\text{your features}} \cdot R \implies |H|^2 = \frac{1}{|Q_1|^2 \dots |Q_K|^2 |R|^2}$$

Wherever a factor has roots near the unit circle, the envelope must peak.

standard prior $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$



spectral prior $\mathbf{a} \sim \mathcal{N}(\boldsymbol{\mu}_M, \lambda \mathbf{I})$



Predicted envelopes $|H(f)|^2$ under each prior; bold: prior mean; dotted: the three resonances given to the spectral prior. More examples behind the QR code.

THE IDEA

AR(4), impose one resonance $Q(z) \mid A(z)$:

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$

$$Q(z) = 1 - cz^{-1} + dz^{-2}, \quad c = 2\rho \cos \omega_0, d = \rho^2$$

Note that *divisibility is linear in a*:

$$Q \mid A \iff \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0}$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it *on average*, $\mathbb{E}_{\mathbf{a}}[\mathbf{F}\mathbf{a} + \mathbf{c}] = \mathbf{0}$, with the minimum- D_{KL} update of $\mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$: the covariance stays put, the mean shifts (λ cancels):

$$\boldsymbol{\mu}_M = -\mathbf{F}^\top (\mathbf{F}\mathbf{F}^\top)^{-1} \mathbf{c}$$

FEATURES

One factor $Q(z)$ per expected feature:

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2}$	2
Real pole at ρ	$1 - \rho z^{-1}$	1
Tilt $-6k$ dB/oct	$(1 - \rho z^{-1})^k$	k
Seasonal pole of period s	$1 - z^{-s}$	s
k -fold resonance	$(1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2})^k$	$2k$

REWARD more stable fits? better-posed problems? we don't know yet: let's find out.

